$y^2dx + (xy+1)dy = 0$

$Q(y) = \frac{1}{M}\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)$

$\lambda = e^{\int Q(y)dy}$

$\frac{\partial M}{\partial y} = 2y \qquad \frac{\partial N}{\partial x} = y$

$Q(y) = \frac{1}{y^2}(y - 2y) = -\frac{1}{y}$

$\lambda = e^{-\int \frac{1}{y} dy} = e^{\ln|y^{-1}|} = \frac{1}{y}$

$y\,dx\,(x + \frac{1}{y})\,dy = 0$

$\frac{\partial u}{\partial x} = y\,dx$

$u = xy + g(y)$

$\frac{\partial u}{\partial y} = x + g'$

$g' = \frac{dy}{y}$

$g = \ln y$

$xy + \ln y = C$

1) $2ydx - xdy = 0$

$\frac{\partial M}{\partial y} = 2 \qquad \frac{\partial N}{\partial x} - 1$

$Q(x) = \frac{1}{-x}(2+1) = \frac{3}{-x}$

$\lambda = e^{-\int \frac{3}{x} dx} = e^{-3\ln|x|}$

$= \frac{1}{x^3}$

$2y\,dx - dy = 0$

$$\frac{\quad}{x^3} \qquad \frac{\quad}{x^2} = 0$$

$$\frac{\partial u}{\partial y} = -\frac{dy}{x^2}$$

$$u = -\frac{y}{x^2} + g(x)$$

$$\frac{\partial u}{\partial x} = \frac{2y}{x^3} + g'$$

$$g' = 0$$

$$-\frac{y}{x^2} = C$$

2) $(3x^2 - y^2)dy - 2xydx = 0$

$$\frac{\partial M}{\partial y} = -2x \qquad \frac{\partial N}{\partial x} = 6x$$

$$Q(y) = \frac{1}{M}\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) = \frac{1}{-2xy}\left(6x + 2x\right) = -\frac{4}{y}$$

$$\lambda = e^{\int -\frac{4}{y}dy} = \frac{1}{y^4}$$

$$\left(\frac{3x^2}{y^4} - \frac{1}{y^2}\right)dy - \frac{2x}{y^3}dx = 0$$

$$\frac{\partial u}{\partial x} = -\frac{2x}{y^3}dx$$

$$u = -\frac{x^2}{y^3} + g(y)$$

$$\frac{\partial u}{\partial y} = 3\frac{x^2}{y^4} + g'$$

$$g' = -\frac{1}{y^2}\,dy$$

$$g = \frac{1}{y}$$

$$-\frac{x^2}{y^3} + \frac{1}{y} = k$$

$$y^2 - x^2 = C$$

7) $xdy + ydx + 3x^3y^4dy = 0$

lo miembro, como consideramos
nte, ha de ser función de $\mu = xy$.

$$Q(\mu) = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{y.N - x.M} \quad (5)$$

$$\boxed{\lambda = e^{\int Q(\mu)du}}$$

$$(x + 3x^3y^4)\,dy + y\,dx = 0$$

$$\frac{\partial M}{\partial y} = 1 \qquad \frac{\partial N}{\partial x} = 1 + 9x^2y^4$$

$$Q(u) = \frac{\cancel{1} - \cancel{1} - 9x^2y^4}{y(x + 3x^3y^4) - xy} = \frac{-9x^2y^4}{\cancel{xy} + 3x^3y^5 - \cancel{xy}} = \frac{-9\cancel{x^2}\cancel{y^4}}{3\cancel{x^3}\cancel{y^5}}$$

$$= \frac{-3}{xy}$$

$$\lambda = e^{\int -\frac{3}{u}du}$$

$$u = xy$$

$$\lambda = \frac{1}{u^3} = \frac{1}{(xy)}$$

$$\left(\frac{1}{y} + 3x^2y^3\right)dy + \frac{dx}{x} = 0$$

$$\frac{\partial U}{\partial x} = \frac{dx}{x}$$

$$U = \ln x + g(y)$$

$$u = \ln x + g(y)$$

$$\frac{\partial u}{\partial y} = g'$$

$$\left(\frac{1}{y} + 3x^2y^3\right) dy = g'$$

$$\ln y + \frac{3}{4}x^2y^4 = g$$

$$\ln x + \ln y + \frac{3}{4}x^2y^4 = k$$

$$\ln(xy)^4 + 3x^2y^4 = C$$